

- Definition of t test: “ It’s a method of testing hypothesis about the mean of small sample drawn from a normally distributed population when the standard deviation for the sample is unknown.”

Both t-tests and z-tests are used to compare means and test hypotheses, but they have their unique advantages and limitations:

### **t-Test**

Advantages:

1. Small Sample Size: Ideal for small sample sizes (typically  $n < 30$ ).
2. Unknown Population Standard Deviation: Used when the population standard deviation is unknown and estimated from the sample.
3. Flexibility: Can be used for paired samples, independent samples, and one-sample tests.

Limitations:

1. Less Robust with Large Samples: Although it can be used, it becomes less efficient compared to a z-test as sample sizes increase.
2. Normality Assumption: Assumes the sample data is approximately normally distributed, which might not hold true for very small samples.

### **z-Test**

Advantages:

1. Large Sample Size: Ideal for large sample sizes (typically  $n > 30$ ).
2. Known Population Standard Deviation: More efficient when the population standard deviation is known.
3. Simplicity: Computationally simpler when conditions for its use are met.

Limitations:

1. Not Suitable for Small Samples: Less reliable for small sample sizes.
2. Requires Known Population Standard Deviation: Not applicable when the population standard deviation is unknown and has to be estimated from the sample.

In summary, use a t-test when dealing with small sample sizes or unknown population standard deviations, and a z-test for large sample sizes with known population standard deviations. Clear which one you need to use for your data?

### **Assumptions of t-Test**

Dependent variables are interval or ratio.

The population from which samples are drawn is normally distributed.

Samples are randomly selected.

The groups have equal variance (Homogeneity of variance).

The t-statistic is robust (it is reasonably reliable even if assumptions are not fully met.)

### **Applications of t test**

The calculation of a confidence interval for a sample mean.

To test whether a sample mean is different from a hypothesized value.

To compare mean of two samples.

To compare two sample means by group.

## Types of "t" test

**Single sample t-test** — we have only 1 group; want to test against a hypothetical mean.

**Independent samples t-test** — we have 2 means, 2 groups; no relation between groups, Eg: When we want to compare the mean of T/T group with Placebo group.

**Paired t-test** — It consists of samples of matched pairs of similar units or one group of units tested twice. Eg: Difference of mean pre & post drug intervention.

## Single Sample t-test

It is used in measuring whether a sample value significantly differs from a hypothesized value. For example, a research scholar might hypothesize that on an average it takes 3 minutes for people to drink a standard cup of coffee.

He conducts an experiment and measures how long it takes his subjects to drink a standard cup of coffee.

The one sample t-test measures whether the mean amount of time it took the experimental group to complete the task varies significantly from the hypothesized 3 minutes value.

## Equation for a one-sample t-test

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where

$t$  = the  $t$  statistic

$\bar{x}$  = the mean of the sample

$\mu$  = the comparison mean

$s$  = the sample standard deviation

$n$  = the sample size

## Example

67, 67, 67, 68, 66, 67, 66, 67, 67, 70, 69, 66, 66, 67, 68, 67, 67, 68, 68, 71

Mean  $\bar{X}$  = 67.45  $\sigma_s$  = 1.32

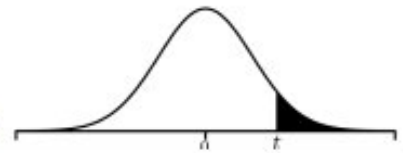
Is the mean different from population mean  $\mu$  = 67.39

$$t = \frac{67.45 - 67.39}{1.32/\sqrt{20}} = 0.20$$

For 19 df critical value of  $t$  for significance level 0.05 is 1.73

Since  $0.20 < 1.73$ , The score has not improved is **not rejected**.

## Critical Values for Student's *t*-Distribution



| df | Upper Tail Probability: $\Pr(T > t)$ |       |        |        | df | Upper Tail Probability: $\Pr(T > t)$ |       |       |       |
|----|--------------------------------------|-------|--------|--------|----|--------------------------------------|-------|-------|-------|
|    | 0.1                                  | 0.05  | 0.02   | 0.01   |    | 0.1                                  | 0.05  | 0.02  | 0.01  |
| 1  | 3.078                                | 6.314 | 15.895 | 31.821 | 11 | 1.363                                | 1.796 | 2.328 | 2.718 |
| 2  | 1.886                                | 2.920 | 4.849  | 6.965  | 12 | 1.356                                | 1.782 | 2.303 | 2.681 |
| 3  | 1.638                                | 2.353 | 3.482  | 4.541  | 13 | 1.350                                | 1.771 | 2.282 | 2.650 |
| 4  | 1.533                                | 2.132 | 2.999  | 3.747  | 14 | 1.345                                | 1.761 | 2.264 | 2.624 |
| 5  | 1.476                                | 2.015 | 2.757  | 3.365  | 15 | 1.341                                | 1.753 | 2.249 | 2.602 |
| 6  | 1.440                                | 1.943 | 2.612  | 3.143  | 16 | 1.337                                | 1.746 | 2.235 | 2.583 |
| 7  | 1.415                                | 1.895 | 2.517  | 2.998  | 17 | 1.333                                | 1.740 | 2.224 | 2.567 |
| 8  | 1.397                                | 1.860 | 2.449  | 2.896  | 18 | 1.330                                | 1.734 | 2.214 | 2.552 |
| 9  | 1.383                                | 1.833 | 2.398  | 2.821  | 19 | 1.328                                | 1.729 | 2.205 | 2.539 |
| 10 | 1.372                                | 1.812 | 2.359  | 2.764  | 20 | 1.325                                | 1.725 | 2.197 | 2.528 |

### Example 2:

70      69      66      69      70      67      69      68      70      69      68

68.64

1.29

Is the mean different from population mean  $\mu = 67.39$

$$t = \frac{68.64 - 67.39}{1.29/\sqrt{10}} = 3.06$$

For 9 df critical value of *t* for significance level 0.05 is 1.83

Since  $3.06 > 1.73$ , The score has not improved is **rejected**.

## Independent samples t-test

It is also called unpaired t-test/ two sample t test.

It is the t-test to be used when two separate independent and identically distributed variables are measured.

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Eg:

1. Comparison of quality of life improved for patients who took drug Valporate as opposed to patients who took drug Levetiracetam in myoclonic seizures.
2. Comparison of mean cholesterol levels in treatment group with placebo group after administration of test drug.

78, 82, 87, 65, 75, 82, 71       $\bar{x}_1 = 77.14$      $\sigma_1 = 7.47$

87, 82, 86, 95, 73                       $\bar{x}_2 = 84.60$      $\sigma_2 = 8.02$

Calculate t-value =  $7.46 / 4.565 = 1.633$  (p value is coming to 0.07)

For the .05 probability level with 10 degrees of freedom,  
the critical t score is 2.228 for a two-tailed test and 1.812 for a one-tailed test.

Since the calculated t score is lower than the critical t-score, improvement if any is not significant at the .05 probability level. Cd

### Paired t test

Eg: I. pre-test/post-test samples I n which a factor is measured before and after an intervention,

° Suppose a sample of “n” subjects were given an antihypertensive drug we want to check blood pressure before and after treatment. We want to find out the effectiveness of the treatment by comparing mean pre & post t/t.

' To test the null hypothesis that the true mean difference is zero, the procedure is as follows:

1. Calculate the difference ( $d_i = y_i - x_i$ ) between the two observations on each pair.
2. Calculate the mean difference, d and standard deviation  $\sigma_d$
3. Calculate the standard error of the mean differences.  $e = \sigma_d / \sqrt{n}$
4. Calculate the t-statistic, which is given by  $T = d/e$ , Under the null hypothesis, this statistic follows a t-distribution with  $n - 1$  degrees of freedom.
5. Use tables of the t-distribution to compare your value for T to the t distribution.

|       |    |    |    |    |    |    |    |       |       |       |       |
|-------|----|----|----|----|----|----|----|-------|-------|-------|-------|
| $x_i$ | 78 | 82 | 87 | 65 | 75 | 82 | 71 |       |       |       |       |
| $y_i$ | 82 | 84 | 86 | 72 | 81 | 85 | 72 | d     | SD    | SE    | T     |
| $d_i$ | 4  | 2  | -1 | 7  | 6  | 3  | 1  | 3.143 | 2.795 | 1.056 | 2.976 |

t-value at 0.05 significance level is 1.833

Since  $2.976 > 1.833$ , or corresponding p-value  $0.012 < 0.05$ , null hypothesis is rejected.